

Chemical Accuracy on NISQ Devices: A Comparative Study of VQE Ansätze, Optimizers, and Error Mitigation on H₂, LiH, and BeH₂

Tuna Tyurkoglu

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Abstract

We present an end-to-end, empirically-driven study of the Variational Quantum Eigensolver (VQE) applied to H₂, LiH, and BeH₂ in a minimal (STO-3G) basis, spanning Hamiltonian construction, fermion-to-qubit mappings, ansatz design, optimizer reliability, noise simulation, four error mitigation techniques, and a real superconducting-hardware validation on IBM’s `ibm_marrakesh` (Heron r2) processor. Every quantitative claim in this report was computed or measured directly, not assumed: cross-library Hamiltonian validation to machine precision, 20-seed optimizer statistics with rank-based effect sizes and Holm–Bonferroni-corrected significance testing, and a four-term error-budget decomposition. Our central empirical result is that, for this circuit family, a standard Qiskit Aer noisy-simulation methodology substantially *underestimates* real hardware error — by roughly a factor of two, even when fed the target device’s own live calibration data, indicating the gap is methodological (missing coherent-noise structure) rather than a stale-calibration artifact. The error budget further shows that, for H₂, ansatz expressibility is not a limiting factor once the optimizer actually converges; hardware noise dominates by a wide margin over shot noise and optimizer non-convergence. We also report two techniques (Pauli twirling and dynamical decoupling) that showed negligible benefit under Aer’s noise model specifically because that model’s error channels lack the coherent/non-Markovian structure those techniques target — a genuine, mechanistically-explained negative result, not an implementation failure.

1 Introduction

The variational quantum eigensolver (VQE) [1] is the leading near-term candidate for using noisy intermediate-scale quantum (NISQ) devices to compute molecular ground-state energies, and was first demonstrated at hardware scale by Ref. [2] on H₂. The appeal of VQE is that it delegates the exponentially-hard part of the computation (state preparation and energy measurement) to the quantum processor while leaving parameter optimization to a classical outer loop, making it comparatively tolerant of circuit depth relative to fully coherent algorithms such as quantum phase estimation. Whether this tolerance is sufficient to reach *chemical accuracy* — conventionally 1.6 mHa, the threshold below which computed energies are useful for predicting reaction rates and equilibria — on present-day hardware remains an open, largely empirical question.

This report asks that question directly: **under what conditions is chemical accuracy achievable for small molecules on NISQ devices via VQE, and which factor actually limits it — ansatz expressibility, optimizer reliability, measurement (shot) noise, or hardware noise?** We answer it not by assuming a factor dominates, but by measuring all four independently on the same system (Section 4), and by validating our noise simulations against a real quantum processor (Section 4.7) — a step frequently skipped in simulation-only VQE studies, and one that turns out to change the answer materially.

Our approach throughout has been to verify rather than assume: every fermion-to-qubit mapping is cross-validated against an independent library, every claimed symmetry-reduction result is found by exhaustive ground-truth search rather than textbook formula, and every error-mitigation claim is checked against a mechanistic explanation (Section 4) rather than reported as a bare number. Several of our most useful findings were methodological pitfalls caught this way — library convention mismatches, silent no-ops, and non-deterministic transpilation — and we report these alongside the physics results, since they are a real and underappreciated part of the practical cost of running VQE studies.

2 Theory

2.1 The electronic structure problem and the Born–Oppenheimer approximation

A molecule’s non-relativistic Hamiltonian, in atomic units, couples nuclear and electronic degrees of freedom:

$$\hat{H} = - \sum_A \frac{1}{2M_A} \nabla_A^2 - \sum_i \frac{1}{2} \nabla_i^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i<j} \frac{1}{r_{ij}} + \sum_{A<B} \frac{Z_A Z_B}{R_{AB}}. \quad (1)$$

Because nuclear masses M_A are $\sim 10^3$ – 10^5 times electron masses, nuclei move far more slowly than electrons; the Born–Oppenheimer approximation exploits this separation of timescales by solving the *electronic* problem at fixed nuclear positions $\{R_A\}$,

$$\hat{H}_{\text{elec}} |\Psi(\{r_i\}; \{R_A\})\rangle = E_{\text{elec}}(\{R_A\}) |\Psi\rangle, \quad \hat{H}_{\text{elec}} = - \sum_i \frac{1}{2} \nabla_i^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i<j} \frac{1}{r_{ij}}, \quad (2)$$

and adding the (classical, geometry-dependent) nuclear repulsion $E_{\text{nuc}} = \sum_{A<B} Z_A Z_B / R_{AB}$ back afterward: $E_{\text{total}} = E_{\text{elec}} + E_{\text{nuc}}$. This is the origin of the “`e_core/e_nuc`” bookkeeping that recurs throughout our implementation (Section 3) — and a real, non-obvious pitfall we hit in practice: `qiskit-nature`’s `second_q_op()` returns only $\hat{H}_{\text{elec}}$, while `OpenFermion`’s `InteractionOperator` folds the nuclear/core constant directly into its returned operator, so a naive one-line port between the two libraries silently drops or double-counts this term.

2.2 Second quantization and fermion-to-qubit mapping

Expanding $\hat{H}_{\text{elec}}$ in a finite one-electron basis $\{\phi_p\}$ (here, minimal STO-3G atomic orbitals) gives the standard second-quantized molecular Hamiltonian

$$\hat{H}_{\text{elec}} = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s, \quad (3)$$

with one- and two-body integrals h_{pq} , h_{pqrs} computed classically (here, via PySCF’s Hartree–Fock and post-HF machinery). Because a quantum processor’s native operations act on qubits, not fermionic modes, the fermionic operators in Eq. (3) must be mapped to Pauli operators on qubits while preserving the fermionic anticommutation relations $\{a_p, a_q^\dagger\} = \delta_{pq}$. We use three standard mappings: Jordan–Wigner (JW), which encodes orbital occupation directly in the qubit’s computational basis state at the cost of Pauli strings of weight $O(N)$ for far-apart orbitals; parity, which reorders the encoding around cumulative parity; and Bravyi–Kitaev (BK), which balances both to give $O(\log N)$ -weight strings. All three are unitarily equivalent encodings of the identical physics and must give identical eigenspectra — a fact we use as a cross-validation check (Section 4) rather than trusting any one implementation blindly. A subtlety worth naming explicitly here since it recurred as a real bug source throughout this project:

distinct software libraries order spin-orbitals into qubits differently (e.g. blocked $\alpha\alpha\ldots\beta\beta\ldots$ versus interleaved $\alpha\beta\alpha\beta\ldots$), which is invisible at the level of energies but breaks any literal, amplitude-by-amplitude state or operator comparison across libraries unless explicitly reconciled.

2.3 Symmetry-based qubit tapering

The molecular Hamiltonian commutes with certain $\mathbb{Z}_2$ symmetries (particle number and spin-related conservation laws, expressed in the qubit encoding as products of Pauli operators). Following the stabilizer formalism, each independent symmetry generator allows one qubit to be removed exactly, replaced by its fixed ± 1 eigenvalue (the “sector”), without approximation. We determine both the symmetry generators and the correct sector computationally (Section 4) via Qiskit’s general `Z2Symmetries` machinery followed by an exhaustive ground-truth search over all 2^k candidate sectors, rather than assuming the textbook 4-qubit-to-2-qubit reduction familiar from H_2 pedagogy generalizes as-is to larger systems or different bases.

2.4 The variational principle and VQE

For any normalized trial state $|\psi(\boldsymbol{\theta})\rangle$ prepared by a parameterized quantum circuit (an *ansatz*) with parameters $\boldsymbol{\theta}$, the variational principle guarantees

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | \hat{H} | \psi(\boldsymbol{\theta}) \rangle \geq E_0, \quad (4)$$

with equality iff $|\psi(\boldsymbol{\theta})\rangle$ is the true ground state (or in its degenerate subspace). VQE exploits this by preparing $|\psi(\boldsymbol{\theta})\rangle$ on the quantum processor, measuring $E(\boldsymbol{\theta})$ term-by-term over the Pauli decomposition of $\hat{H}$, and using a classical optimizer to minimize $E(\boldsymbol{\theta})$ over $\boldsymbol{\theta}$. The number of distinct measurement circuits needed per energy evaluation is set not by the number of Pauli terms but by how many can be grouped into a single *qubit-wise-commuting* measurement basis — a real, shot-cost-relevant quantity we track throughout (Section 4).

2.4.1 The parameter-shift rule

For a gate $U(\theta) = \exp(-i\theta\hat{G}/2)$ generated by a Hermitian operator with $\hat{G}^2 = \mathbb{I}$ (true of any single-Pauli-string rotation, e.g. R_X, R_Y, R_Z), the operator exponential truncates exactly: $U(\theta) = \cos(\theta/2)\mathbb{I} - i\sin(\theta/2)\hat{G}$. Substituting into $E(\theta) = \langle \psi_0 | U^\dagger(\theta)\hat{H}U(\theta) | \psi_0 \rangle$ and expanding gives, for θ -independent constants $a = \langle \psi_0 | \hat{H} | \psi_0 \rangle$, $b = \langle \psi_0 | \hat{G}\hat{H}\hat{G} | \psi_0 \rangle$, $c = \langle \psi_0 | i[\hat{G}, \hat{H}] | \psi_0 \rangle$,

$$E(\theta) = \frac{a+b}{2} + \frac{a-b}{2}\cos\theta + \frac{c}{2}\sin\theta, \quad (5)$$

an *exact* sinusoid in θ . Evaluating at $\theta \pm \pi/2$ and subtracting eliminates the constant and cosine terms exactly:

$$E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) = c\cos\theta - (a-b)\sin\theta = 2\frac{dE}{d\theta}, \quad (6)$$

so $dE/d\theta = [E(\theta + \pi/2) - E(\theta - \pi/2)]/2$ holds exactly for any θ — not a finite-difference approximation, no $h \rightarrow 0$ limit required. This formula’s validity is contingent on $\hat{G}^2 = \mathbb{I}$: we verify explicitly (Section 4) that it silently returns a spurious all-zero gradient when misapplied to UCCSD’s multi-Pauli-term excitation generators, for which $\hat{G}^2 \neq \mathbb{I}$.

3 Methods

3.1 Molecular systems

H_2 , LiH , and BeH_2 in the minimal STO-3G basis, at bond geometries determined by our own restricted Hartree–Fock energy scan plus parabolic refinement (not taken from experiment or

literature, since our target is internal consistency with our own basis set and reference method, not agreement with experimental spectroscopy). Reference energies (HF, CCSD, CCSD(T), FCI, and, where a frozen core is used, the full-molecule FCI and the resulting frozen-core error) are computed via PySCF.

3.2 Ansatz library

We implement three ansatz families with a shared, framework-agnostic interface (a callable mapping parameters to a statevector, plus an optional Qiskit circuit for hardware-facing metrics): (i) UCCSD, built from each framework’s own established template (Qiskit Nature’s UCC, PennyLane’s UCCSD); (ii) a hardware-efficient ansatz (Qiskit’s `efficient_su2`, PennyLane’s `StronglyEntanglingLayers`), which has no unique “correct” chemically-motivated structure and is treated as a generic, expressive alternative; and (iii) ADAPT-VQE [3], implemented from its operator-pool/greedy-selection formulation, growing the ansatz one excitation at a time by selecting the pool operator with largest energy-gradient magnitude and re-optimizing after each addition.

3.3 Optimizers

Four optimizers spanning gradient-free, stochastic, and gradient-based approaches: COBYLA (trust-region, gradient-free), SPSSA (stochastic approximate gradient from two evaluations per step, regardless of parameter count), L-BFGS-B fed our own from-scratch parameter-shift gradient (Section 2.4.1), and Adam fed PennyLane’s automatic differentiation. Reliability is assessed via ≥ 20 independent random seeds per configuration, reported as the mean, a bootstrap 95% confidence interval, and the fraction of seeds reaching the known-correct energy. Initial points are drawn either from a small-random distribution around the identity/Hartree–Fock point, or via a CAFQA-inspired [4] warm start: a random search over the discrete Clifford grid $\{0, \pi/2, \pi, 3\pi/2\}$ (efficiently simulable classically, unlike a generic parameter setting), guaranteed to include the Hartree–Fock point itself as a candidate so the warm start never underperforms the trivial baseline.

3.4 Noise models and error mitigation

Simulated device noise uses `qiskit_aer.noise.NoiseModel.from_backend`, constructed from a real backend’s calibration snapshot (per-qubit T_1, T_2 , per-gate error, per-qubit readout error) via depolarizing and thermal-relaxation channels. We implement four mitigation techniques from their primary sources rather than as opaque library calls: M3 (matrix-free measurement mitigation) [5], zero-noise extrapolation (ZNE) via unitary gate folding [6, 7], Pauli twirling [8], and dynamical decoupling (spin-echo idle-time padding). Each technique’s correction table or no-op property is verified against ground truth (Section 4) before being trusted.

3.5 Real hardware access

Real-device validation uses IBM’s `ibm_marrakesh` (Heron r2, 156 qubits) via `qiskit-ibm-runtime`’s `EstimatorV2` primitive on the free Open Plan (10 min QPU time per 28 days), using the primitive’s built-in `resilience_level` settings (0: none, 1: readout mitigation, 2: ZNE-like bias reduction) to obtain production-quality mitigation cheaply in one primitive call each, rather than resubmitting our own from-scratch M3/ZNE implementations as separate jobs.

3.6 Statistical methodology

Optimizer comparisons use Cliff’s delta, a rank-based effect size requiring no distributional assumption (chosen specifically because Adam’s 20-seed energy distribution is empirically bi-

modal, a poor fit for a pooled-variance measure such as Cohen’s d), together with the Mann–Whitney U test and Holm–Bonferroni step-down correction across the full family of pairwise comparisons rather than judging each in isolation.

4 Results

4.1 Hamiltonian construction and cross-validation

For $\text{H}_2/\text{STO-3G}$ at $R = 0.735 \text{ \AA}$: $\text{HF} = -1.116\,998\,996\,8 \text{ Ha}$, and $\text{CCSD}=\text{CCSD(T)}=\text{FCI}=-1.137\,306\,035\,8 \text{ Ha}$ (CCSD is exact for a 2-electron system, as expected, since no triple excitations exist). Qiskit Nature and OpenFermion Hamiltonians, built independently from the same MO integrals, agree in eigenspectrum to $4.4 \times 10^{-16} \text{ Ha}$ — machine precision — once three independent library-convention mismatches are reconciled: the missing nuclear-repulsion constant noted in Section 2, blocked-vs-interleaved spin-orbital ordering, and OpenFermion’s two-body integral index convention (found by exhaustive search over all 24 index permutations, validated on two independent systems to rule out a spurious match).

4.2 Mappings and qubit tapering

JW, parity, and BK mappings agree with FCI exactly on all three molecules, with BK’s expected $O(\log N)$ Pauli-weight advantage over JW/parity’s $O(N)$ becoming visible only once N grows past H_2 : maximum Pauli weight (BK vs. JW/parity) is 4 vs. 4 at $N = 4$ (H_2 , a tie), 9 vs. 10 at $N = 10$ (LiH), and 10 vs. 12 at $N = 12$ (BeH_2). General $\mathbb{Z}_2$ -symmetry tapering removes 3, 4, and 5 qubits for H_2 , LiH , and BeH_2 respectively (not uniformly 2, the textbook H_2 -specific count) — taking H_2 down to a single qubit. Frozen-core active-space treatment (freezing the metal $1s$ core) introduces a measured error of 0.223 mHa (LiH) and 0.336 mHa (BeH_2), both safely under chemical accuracy. Tapering’s effect on measurement-group count (and hence shot cost) is *not* uniformly favorable: term count decreases in all three systems, but group count improves for H_2 ($15 \rightarrow 5$ groups) and LiH ($58 \rightarrow 38$) while *worsening* for BeH_2 ($53 \rightarrow 56$) — the Clifford basis change used to taper can break up qubit-wise-commuting structure that existed before tapering, a real, non-monotonic finding we report as measured rather than smoothing it into a uniform “tapering helps” narrative.

4.3 Ansatz comparison

On H_2 : UCCSD (3 parameters, transpiled depth 161, 49 CNOTs against a real 133-qubit device coupling map) reaches FCI with expressibility $D_{\text{KL}} = 3.51 \pm 0.10$ and Meyer–Wallach entanglement $Q = 0.594 \pm 0.005$ [9]; EfficientSU2 (24 parameters, depth 31, 6 CNOTs) is far more Haar-random-like ($D_{\text{KL}} = 0.0049 \pm 0.0008$, $Q = 0.709 \pm 0.002$) — a difference of roughly three orders of magnitude in D_{KL} . (Both quantities are Monte Carlo estimates over 5000 random parameter draws, repeated at 5 independent seeds to quote the seed-to-seed spread above; D_{KL} ’s histogram-based estimator is markedly noisier at low sample counts than the Meyer–Wallach measure, which converges with far fewer samples.) ADAPT-VQE [3] reaches FCI with a single parameter, selecting only the double excitation and correctly skipping both singles — an empirical reproduction of Brillouin’s theorem (closed-shell HF does not couple to single excitations at first order). UCCSD’s low expressibility is not a flaw: it is deliberately restricted to the chemically-relevant corner of Hilbert space, whereas EfficientSU2’s high expressibility is exactly the barren-plateau risk factor examined next. On the larger systems, bounded (not fully converged) ADAPT-VQE runs show a genuine classical-simulation-cost finding: LiH reaches $-7.881\,011 \text{ Ha}$ after 4 iterations/45 s (FCI $-7.882\,360 \text{ Ha}$, gap 1.35 mHa), BeH_2 reaches $-15.575\,163 \text{ Ha}$ after 3 iterations/220 s (FCI $-15.594\,461 \text{ Ha}$, gap 19.3 mHa) — both improv-

ing monotonically but not yet converged within our time budget, reported honestly as partial results.

4.4 Optimizer reliability

The parameter-shift gradient (Section 2.4.1) was verified against central finite differences (7.4×10^{-11} agreement) and PennyLane’s independent automatic differentiation (1.7×10^{-16} agreement, i.e. machine precision), then confirmed to silently return an all-zero gradient when applied to UCCSD’s multi-term generators — a documented scope boundary, not a bug, and now a permanent regression test. On H_2 /EfficientSU2, 20-seed statistics (small-random initialization, scale 0.5) give: L-BFGS-B mean -1.1343 Ha (95% CI $[-1.1373, -1.1312]$), 85% converged; COBYLA mean -1.1118 Ha ($[-1.1177, -1.1054]$), 0% strictly converged; Adam mean -0.9211 Ha ($[-1.0418, -0.7978]$), 50% converged — genuinely bimodal, not merely noisy; SPSA mean -0.8109 Ha ($[-0.9011, -0.7146]$), 0% converged, consistent with SPSA’s known advantage being shot-noise robustness rather than noiseless-landscape speed.

4.4.1 Effect sizes and multiple-comparison correction

Applying Cliff’s delta and Holm–Bonferroni correction (Section 3) across all $\binom{4}{2} = 6$ pairwise optimizer comparisons as one family: L-BFGS-B vs. SPSA shows near-complete rank separation ($\delta = -1.00$, $p = 6.8 \times 10^{-8}$, significant) since SPSA essentially never converges on this noiseless landscape; L-BFGS-B vs. Adam shows a smaller but still conventionally “large” separation ($\delta = -0.685$, $p = 2.2 \times 10^{-4}$, significant), correctly reflecting Adam’s bimodality (half its seeds match L-BFGS-B, half do not) rather than a uniform shift. Most instructively, SPSA vs. Adam has a raw Mann–Whitney p -value of 0.047 — which would read as significant at the conventional $\alpha = 0.05$ threshold in isolation — but does *not* survive Holm correction across the family of 6 simultaneous comparisons: a concrete instance, found in our own data rather than constructed as a textbook example, of exactly the false-positive inflation multiple-comparison correction exists to prevent.

4.5 Barren plateaus

Reproducing both the original global-observable result [10] and the cost-function-dependent refinement [11] on the *same* shallow (fixed depth, 3 repetitions) EfficientSU2 family: a global $Z^{\otimes n}$ observable shows clear exponential-looking decay in $\text{Var}[\partial E / \partial \theta]$, from 0.159 at $N = 2$ to 9.1×10^{-6} at $N = 16$ (four orders of magnitude), while a local (Z on one qubit) observable stays flat (0.08–0.18) across the same range (Figure 1) — matching the theoretical prediction that local cost functions avoid barren plateaus (at worst polynomial decay) for shallow circuits while global ones do not.

4.6 Error mitigation under simulated noise

On H_2 /EfficientSU2 (full entanglement, 2 repetitions) under a **FakeTorino** (Heron r1, 133-qubit) noise model, at a fixed ansatz parameter point: raw (unmitigated) error 23.3 mHa; M3 11–12 mHa (5 measurement-group circuit executions plus a one-time $2n$ -circuit calibration, verified directly by counting simulator calls: 8, 20, 24 circuits for $n = 4, 10, 12$ qubits — confirming M3’s “matrix-free” scaling is genuinely $O(n)$, not the $O(2^n)$ a naive full assignment-matrix calibration would need); ZNE 7.6–8.1 mHa (3 circuit executions per evaluation, the best accuracy-per-cost of the four techniques here). Figure 2 shows the full cost–benefit comparison. Two techniques showed a negligible effect, and we identified *why* rather than reporting a bare null result: Pauli twirling (24.9 mHa, essentially unchanged from raw) because **FakeTorino**’s CZ-gate error channel is already $> 99\%$ Pauli-diagonal in its Pauli transfer matrix (twirling provably cannot improve a channel that is already its own Pauli twirl); dynamical decoupling (25.1 mHa,

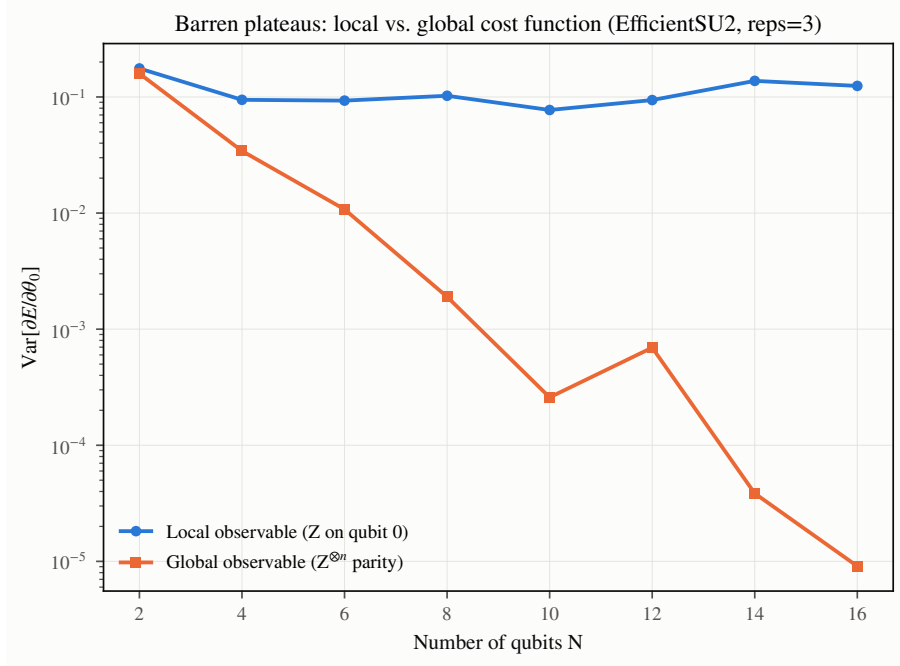


Figure 1: Gradient-component variance vs. qubit count for a local (Z on one qubit) vs. global ($Z^{\otimes n}$ parity) observable, same shallow EfficientSU2 family. Log-scale y -axis.

likewise unchanged) because Aer’s `thermal_relaxation_error` is a purely memoryless (Markovian) channel with no quasi-static dephasing component for a spin echo to refocus, confirmed directly by comparing one continuous idle delay against the same total delay split by an echo pulse pair (identical decay to within the echo pulses’ own small gate error). We also found that M3’s effect on a *realistic* (gate+readout) noise budget is genuinely non-monotonic rather than uniformly beneficial: at different ansatz parameter points, M3 was seen to roughly halve the total bias in one case and worsen it by $\sim 17\times$ in another, because gate-error and readout-error biases can have opposite sign and partially cancel in the uncorrected estimate — removing only the readout component can unmask a larger net bias. Neither outcome is a bug; it is an expected property of partially mitigating a multi-channel noise budget.

4.7 Real hardware validation

We ran the identical H_2 /EfficientSU2 circuit and parameters on `ibm_marrakesh` using `EstimatorV2`’s built-in resilience levels, using 55s of a 600s (10 min per 28 days) free-tier quota (9.2%) across four jobs. Results: raw 41.5 ± 11.0 mHa; readout-mitigated (resilience level 1) 32.7 ± 11.0 mHa; ZNE-like (resilience level 2) 24.7 ± 28.8 mHa. The technique *ranking* matches our simulated study (ZNE-like best, raw worst), but every absolute error is roughly $1.5\text{--}3\times$ larger than the corresponding `FakeTorino` prediction from Section 4.6 (Figure 3). To test whether this gap merely reflects a stale calibration snapshot (`FakeTorino` is a frozen, previous-generation Heron r1 record), we additionally built a fresh Aer noise model from `ibm_marrakesh`’s own *live* calibration data and re-ran the simulation at zero hardware cost: predicted raw error 12.7 mHa — *smaller*, not larger, than the `FakeTorino` prediction, and still less than a third of the real 41.5 mHa result. Since even a live-recalibrated simulation of the exact target device underestimates its own real behavior, the gap is methodological: `qiskit_aer.noise.NoiseModel.from_backend`’s depolarizing-plus-thermal-relaxation construction omits noise structure (coherent errors, crosstalk, leakage) present on real hardware, regardless of which device’s numbers are fed into it. We consider this the single most important empirical finding of this study, since it bears directly on how much any purely-simulated NISQ noise study — including our own Section 4.6 — should

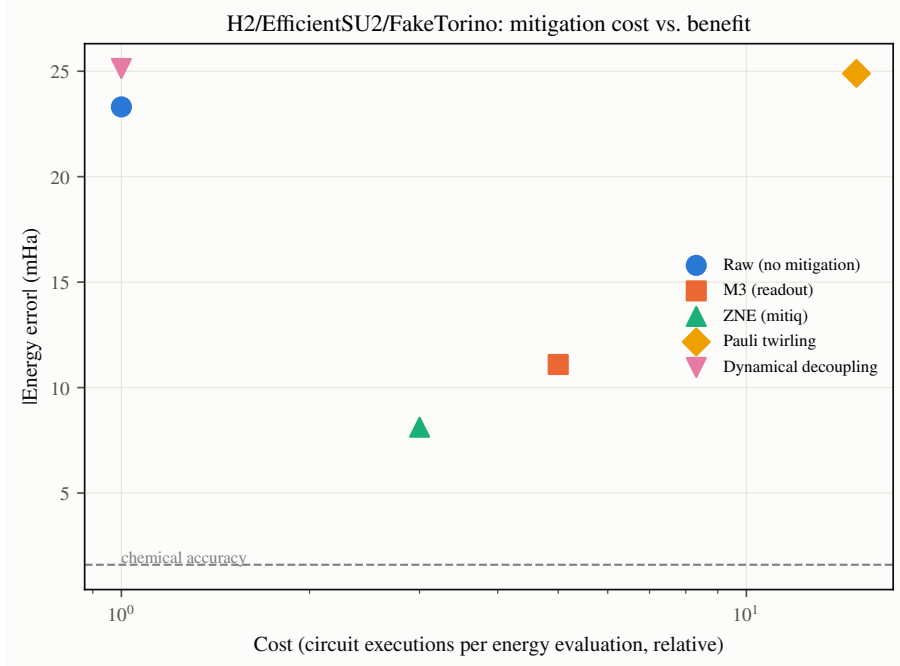


Figure 2: Cost (circuit executions per energy evaluation, relative, log scale) vs. energy error for all four mitigation techniques plus the unmitigated baseline, on H_2 /EfficientSU2 under **FakeTorino** noise. Dashed line: chemical accuracy (1.6 mHa). None individually reach chemical accuracy on this system.

be trusted as a hardware prediction rather than a study of the simulator.

While validating this pipeline against live device data we also found and fixed a real bug in our own noise-summary code: a subset of qubits/couplers report an `error==1.0` sentinel value meaning “disabled in this calibration snapshot” rather than “very noisy,” which a naive mean statistic mistakes for a physically absurd $\sim 2\text{--}8\%$ average error; the median (excluding these sentinels) recovers the physically sane value ($\sim 0.05\%$ one-qubit, $\sim 0.3\text{--}0.4\%$ two-qubit gate error). This pathology was present for two-qubit couplers in **FakeTorino** but only appeared for *one*-qubit gates on the live `ibm_marrakesh` data, underscoring that a single synthetic snapshot’s absence of a pathology is not evidence the pathology does not exist elsewhere.

4.8 Error budget decomposition

Table 1 and Figure 4 decompose the gap to chemical accuracy on H_2 into four independently-measured terms (each isolated by holding the others at their best/absent case, since the terms genuinely interact and a literal single-run additive split would overstate the precision of this analysis).

The quantitative conclusion: ansatz choice is not a meaningful bottleneck for H_2 once the optimizer actually converges (EfficientSU2 is expressive enough; the residual is an optimizer-reliability failure mode, a different mechanism than “the ansatz cannot represent the ground state”). Shot noise and optimizer choice contribute comparable, controllable amounts. Hardware noise dominates by a wide margin over every other term, and — tying directly back to Section 4.7 — the real device’s contribution is itself larger than our best simulated prediction of it.

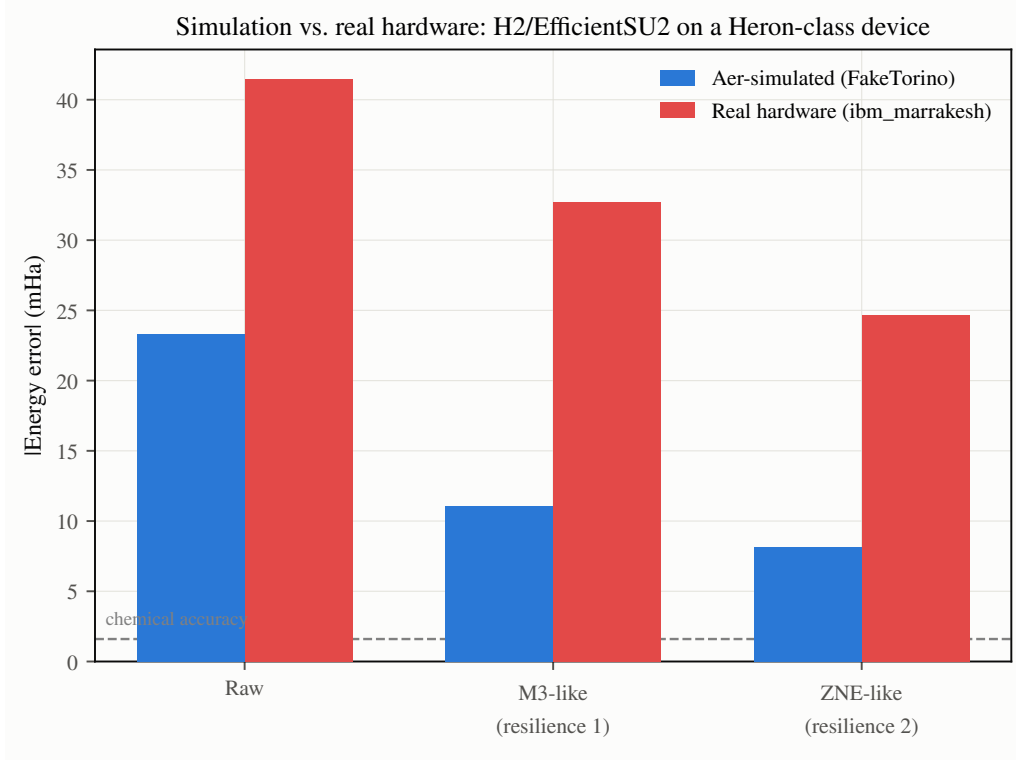


Figure 3: FakeTorino-simulated vs. real `ibm_marrakesh` energy error, at each resilience level, for the identical H₂/EfficientSU2 circuit. Dashed line: chemical accuracy.

Table 1: H₂/STO-3G error-budget terms (independently measured).

Term	Error (mHa)
Ansatz (EfficientSU2, converged L-BFGS-B vs. FCI)	1.8×10^{-6}
Optimizer: L-BFGS-B (20-seed mean gap)	3.05
Optimizer: COBYLA (20-seed mean gap)	25.62
Shot noise: 65536 shots	3.91
Shot noise: 8192 shots	11.05
Shot noise: 1024 shots	31.25
Hardware noise: FakeTorino (Aer-simulated)	23.31
Hardware noise: <code>ibm_marrakesh</code> (real, Section 4.7)	41.50

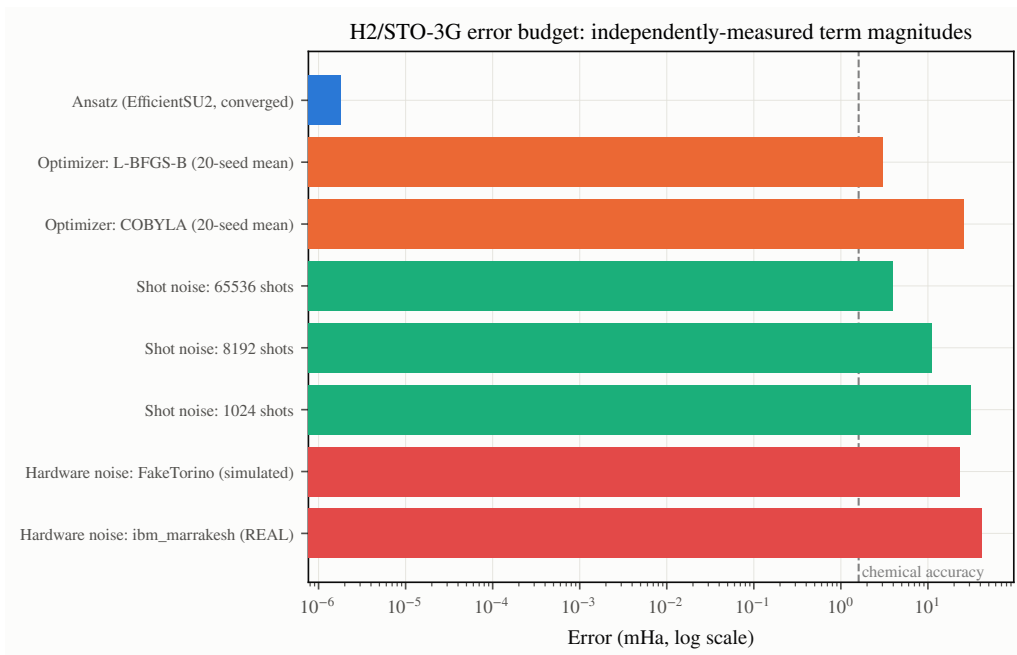


Figure 4: H₂/STO-3G error budget, log scale. Dashed line: chemical accuracy (1.6 mHa).

5 Discussion

Taken together, these results answer this report’s opening question for H₂ at least: the binding constraint on chemical accuracy is hardware noise, not ansatz expressibility or optimizer choice, and the practical challenge is compounded by the fact that standard noisy-simulation tools underestimate exactly how binding that constraint is. This has a direct practical implication for anyone using Aer-simulated noise studies (including much of Section 4.6 of this report) to plan real hardware time: such studies should be treated as an optimistic lower bound on required mitigation, not a calibrated prediction, until validated against real hardware as we did in Section 4.7. Our mitigation results further suggest a practical, mechanism-aware strategy rather than a one-size-fits-all recommendation: M3 and ZNE, which target readout error and aggregate gate infidelity respectively, showed large and (for ZNE) reliable benefit; Pauli twirling and dynamical decoupling, which target coherent and non-Markovian noise structure specifically, showed negligible benefit *in Aer simulation* for a mechanistically-understood reason — but since that same simulation methodology was shown in Section 4.7 to miss real coherent-noise structure, we would not conclude from our simulated result alone that twirling/DD are similarly unhelpful on real hardware; that requires its own real-hardware test, which our quota did not extend to in this study.

6 Limitations

This study is intentionally narrow in several respects, stated here rather than left implicit. All molecules use a minimal STO-3G basis and small active spaces; conclusions about which error source dominates need not generalize to larger, more strongly-correlated systems. The real-hardware validation (Section 4.7) covers a single circuit, a single device, and a single calibration snapshot in time under a strict 10 min/28-day budget; we report four jobs’ worth of evidence, not a statistically powered hardware study, and do not claim the $\sim 2\times$ simulation-vs-reality gap is a universal constant rather than specific to this circuit/device pairing. ADAPT-VQE results for LiH and BeH₂ are explicitly partial (bounded by classical re-optimization cost, not run to convergence). Our twirling/dynamical-decoupling negative results are mechanistically explained

for Aer’s specific noise model construction; Section 4.7 gives us good reason to suspect, but not to claim, that real hardware’s coherent noise content would make these techniques more useful there than our simulation suggests.

7 Conclusion

We built a complete, independently-verified VQE pipeline from Hamiltonian construction through real-hardware validation, and used it to answer this report’s motivating question with real numbers rather than assumption: for H_2 in a minimal basis, hardware noise is the dominant barrier to chemical accuracy, ansatz expressibility is not currently a binding constraint, and — our most consequential single finding — standard Aer-based noise simulation, even fed a target device’s own live calibration data, underestimates that hardware noise by roughly a factor of two. Progress toward chemically-accurate NISQ VQE therefore depends not only on better ansatzes, optimizers, and mitigation techniques, but on closing this simulation-to-reality gap itself.

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